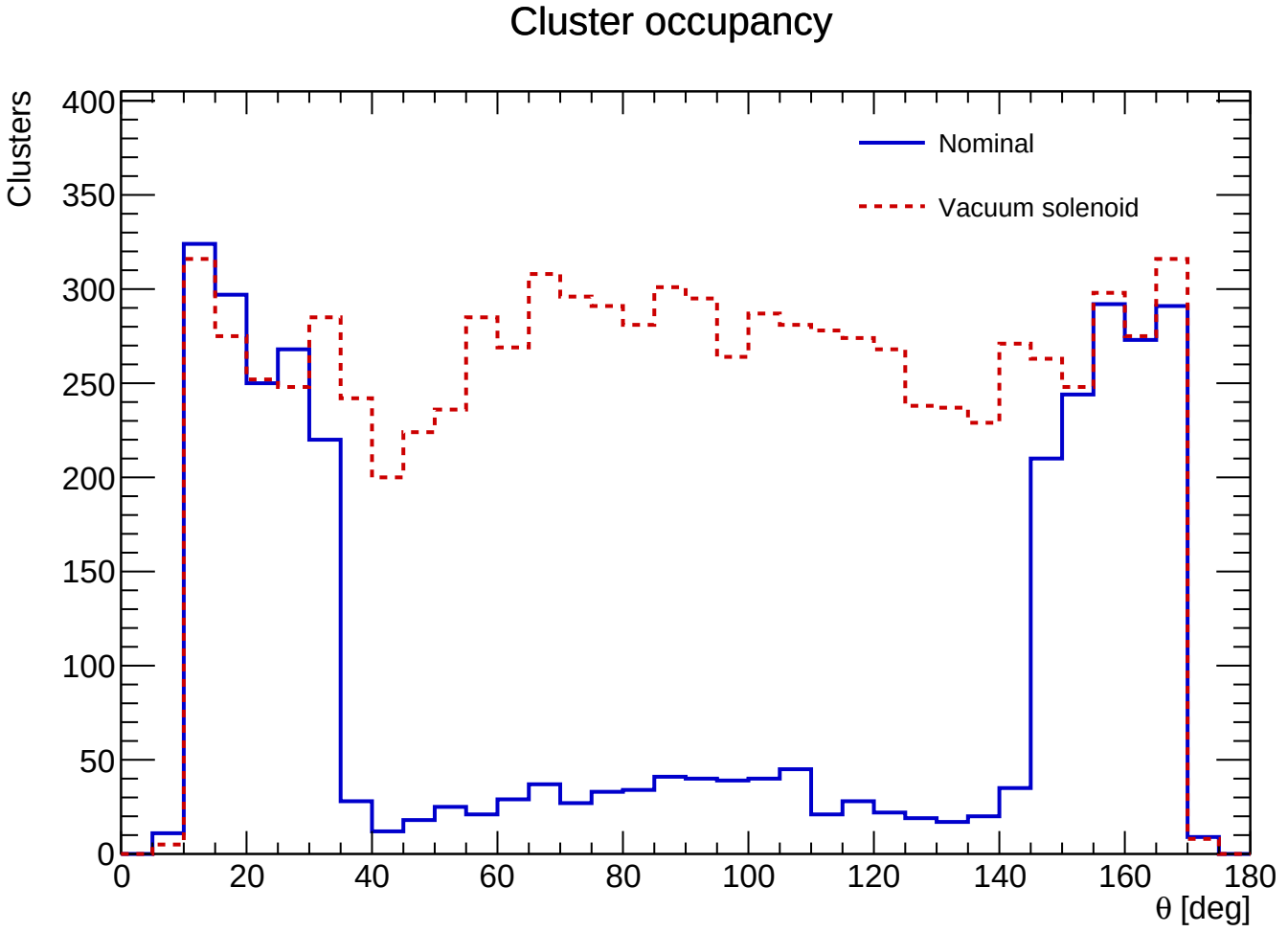
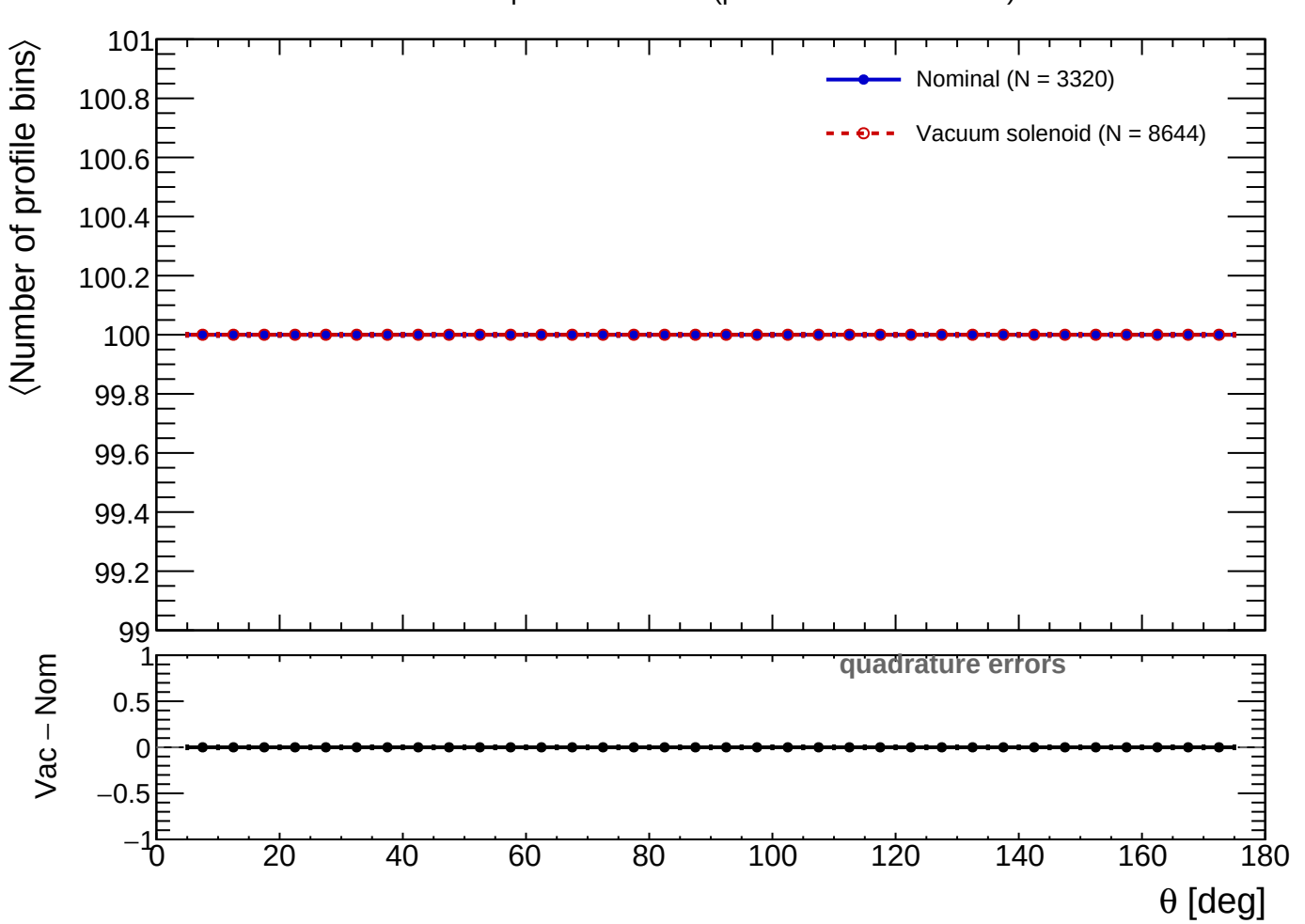
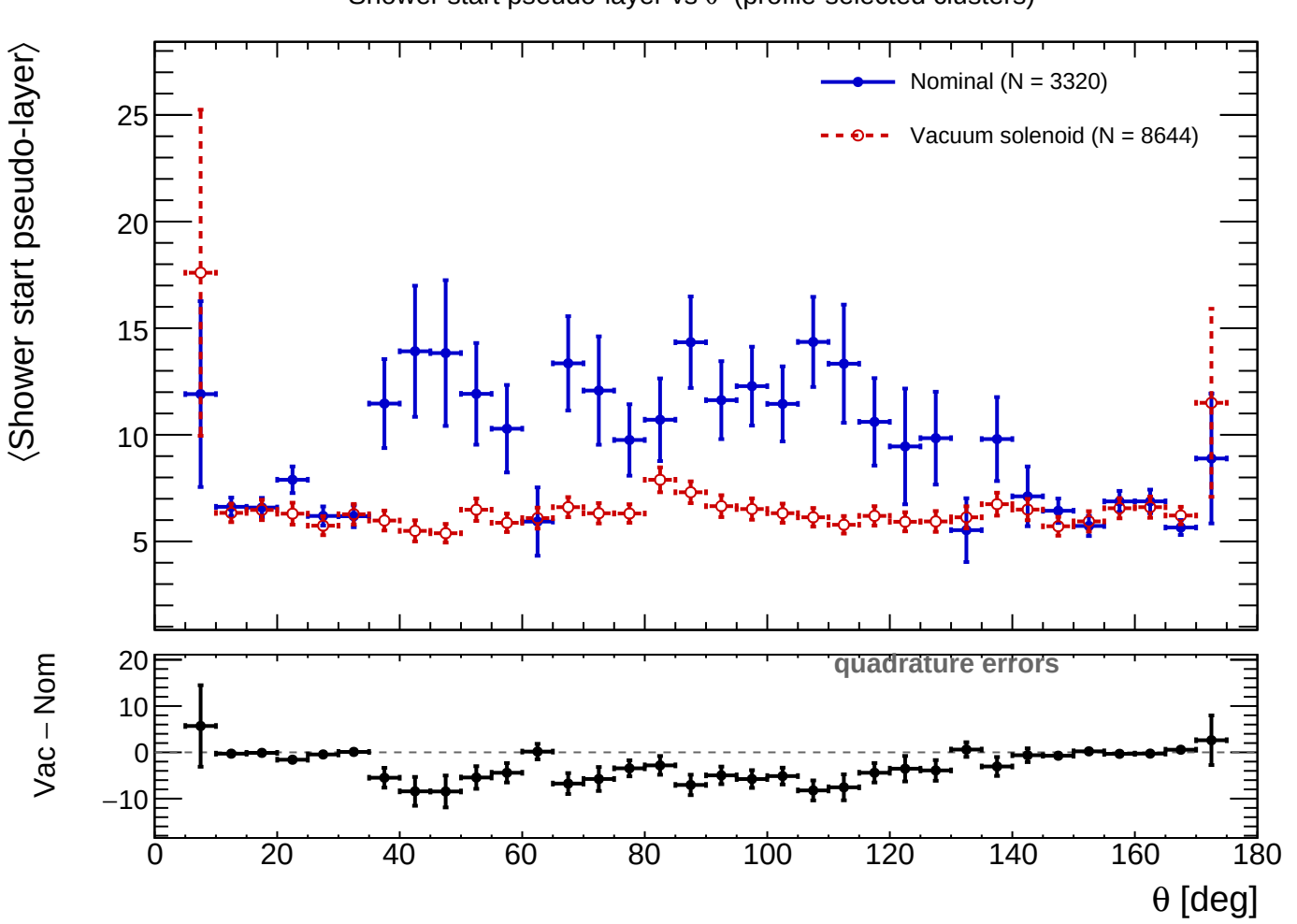
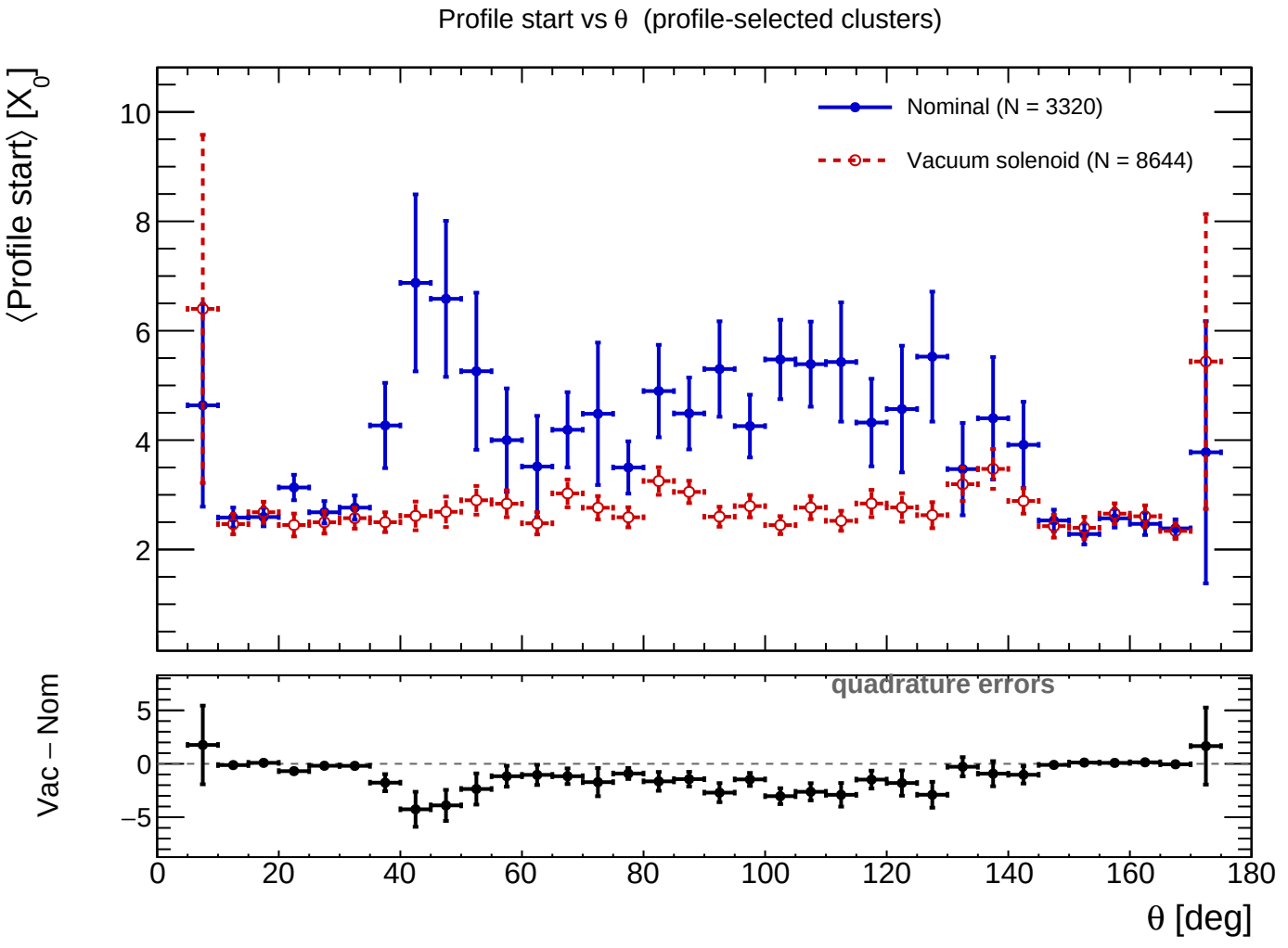
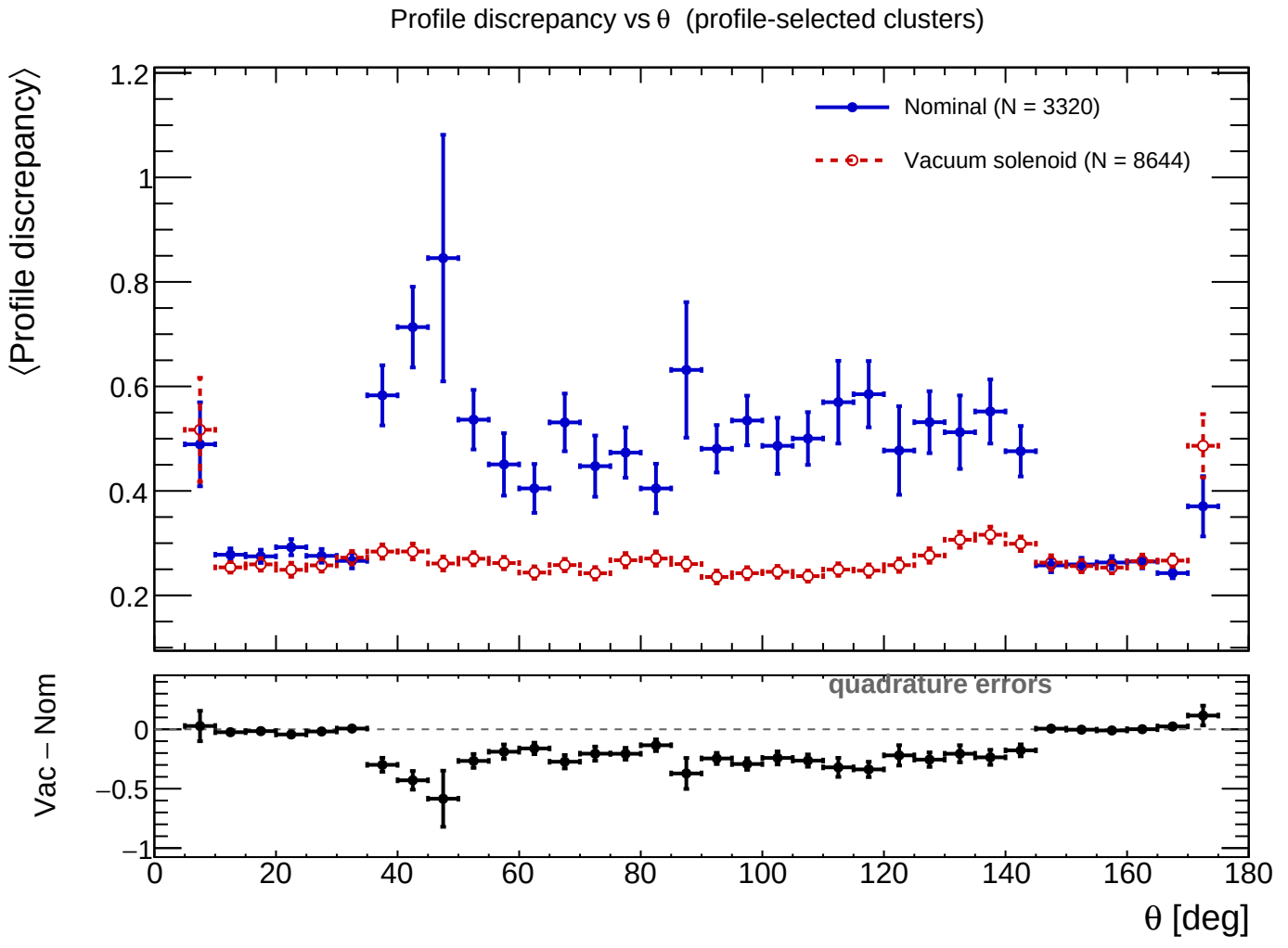
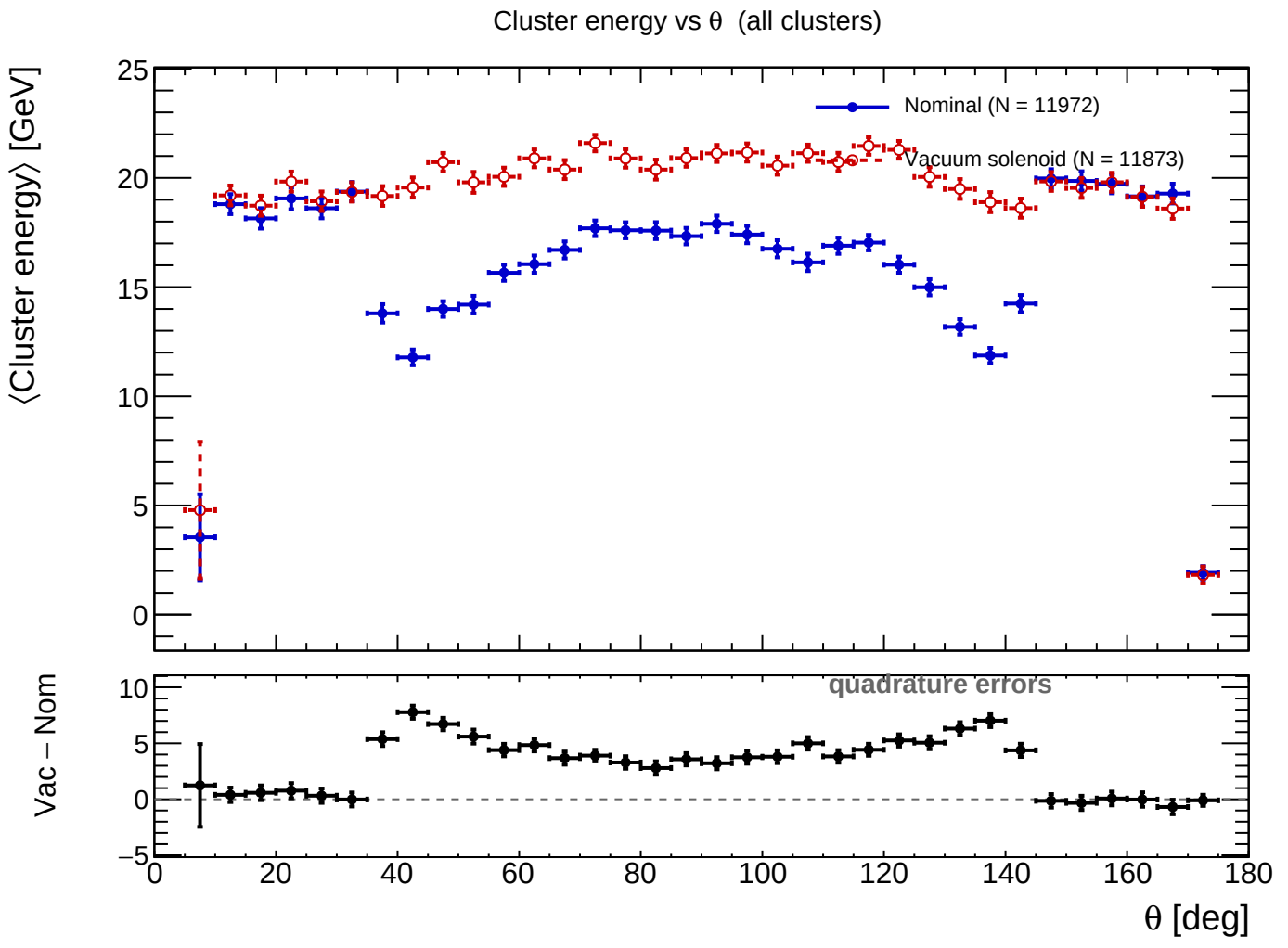

Cluster occupancy
Clusters
400
350
300
250
200
150
100
50
0
0
20
40
60
80
100
120
140
160
180
θ [deg]
Nominal
Vacuum solenoid

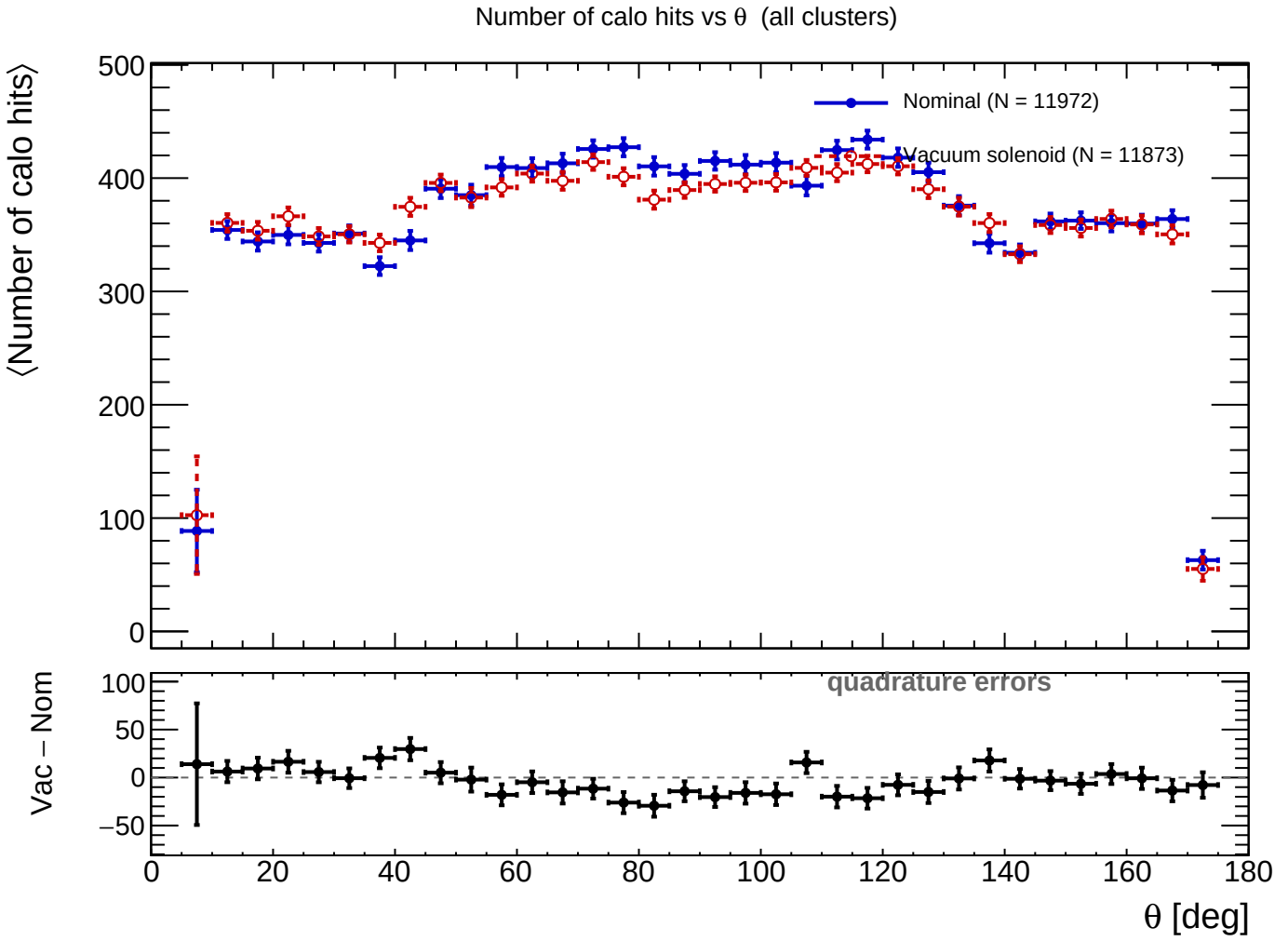

Number of profile bins vs θ (profile-selected clusters)
⟨Number of profile bins⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
101
100.8
100.6
100.4
100.2
100
99.8
99.6
99.4
99.2
99
Vac − Nom
quadrature errors
1
0.5
0
−0.5
−1
0
20
40
60
80
100
120
140
160
180
θ [deg]

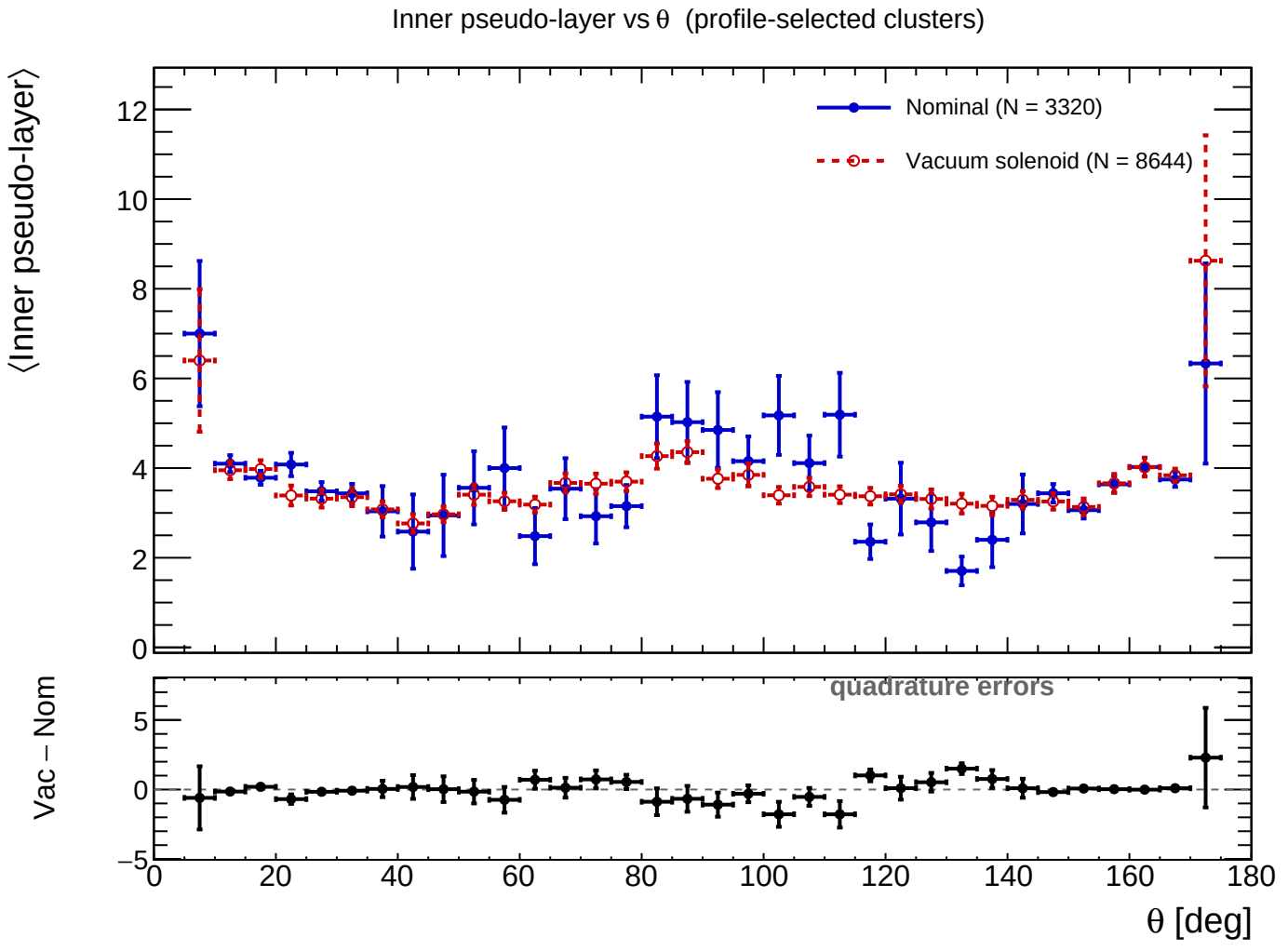

Shower start pseudo-layer vs θ (profile-selected clusters)
⟨Shower start pseudo-layer⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac − Nom
quadrature errors
θ [deg]

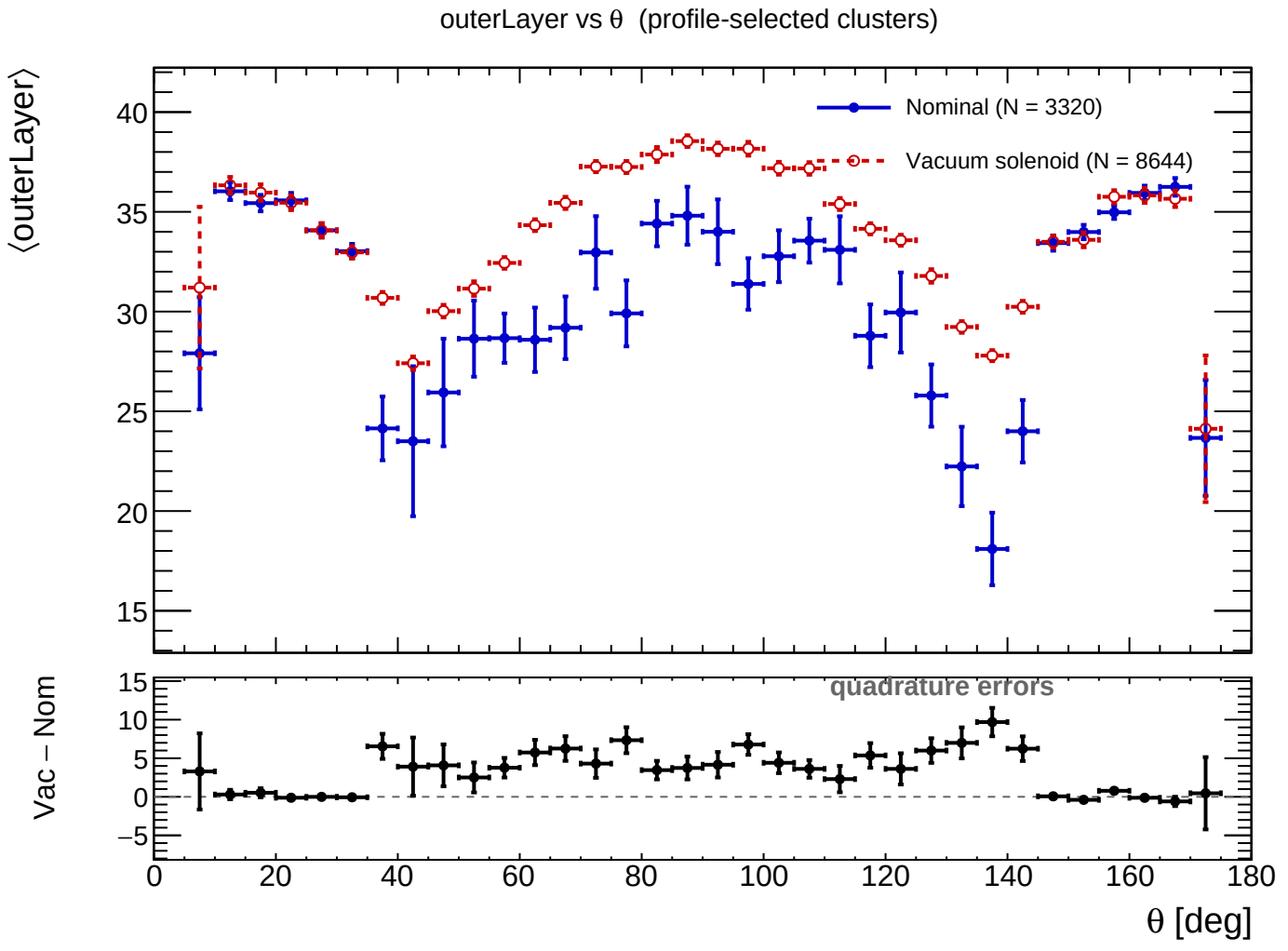

Profile start vs θ (profile-selected clusters)
⟨Profile start⟩ [X0]
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac – Nom
quadrature errors
θ [deg]

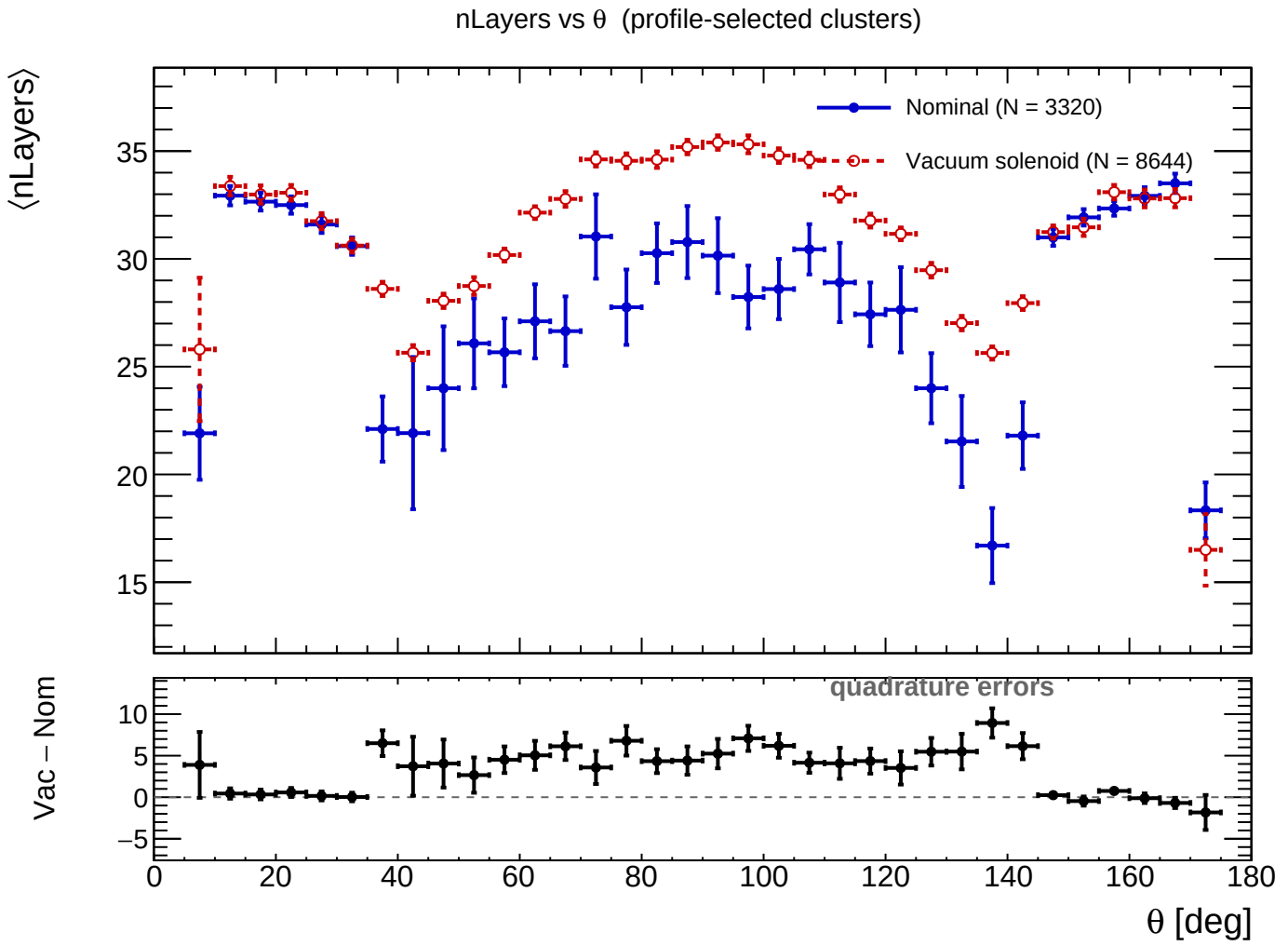

Profile discrepancy vs θ (profile-selected clusters)
⟨Profile discrepancy⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac − Nom
quadrature errors
θ [deg]

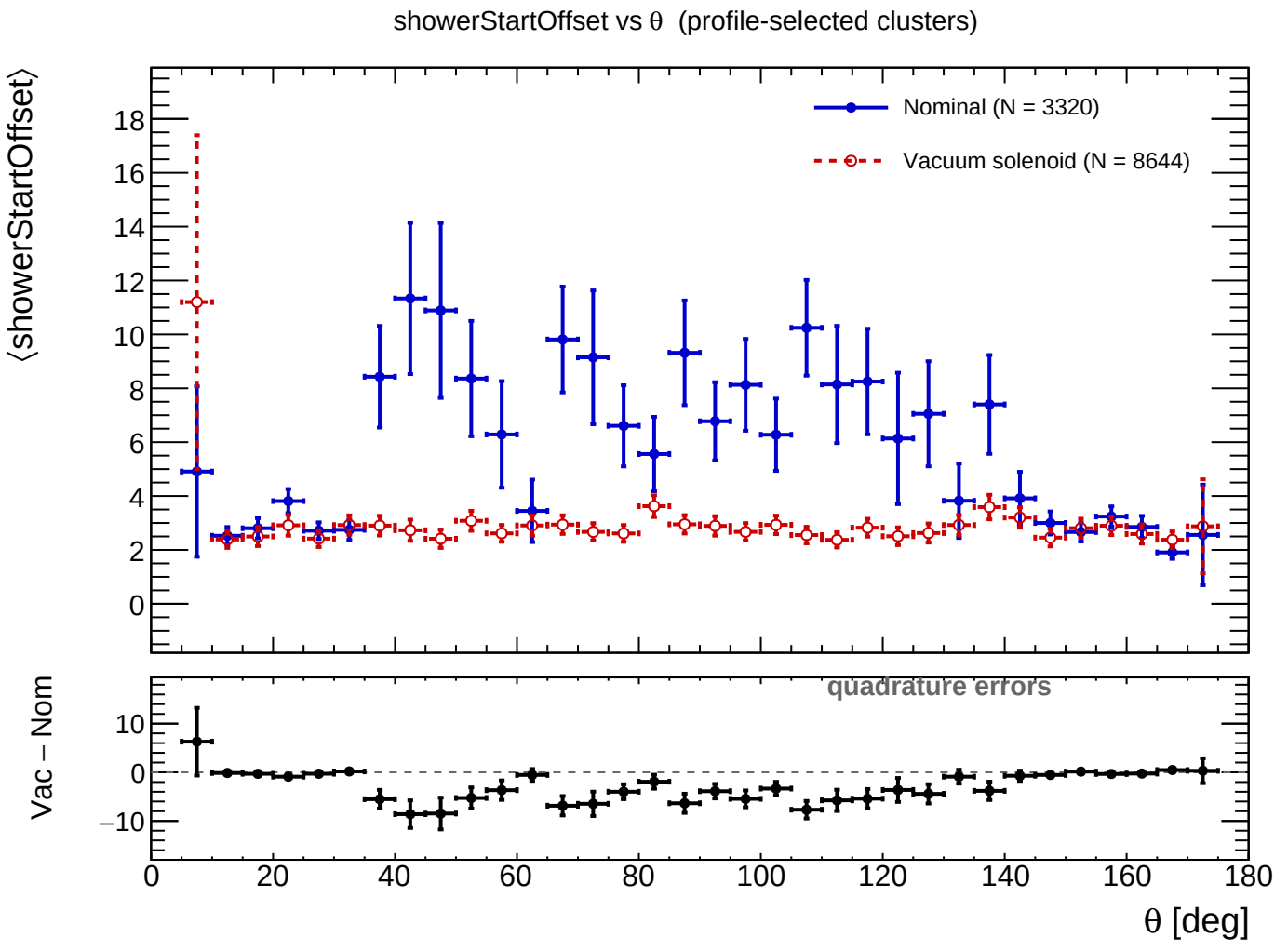

Cluster energy vs θ (all clusters)
⟨Cluster energy⟩ [GeV]
Nominal (N = 11972)
Vacuum solenoid (N = 11873)
Vac – Nom
quadrature errors
θ [deg]

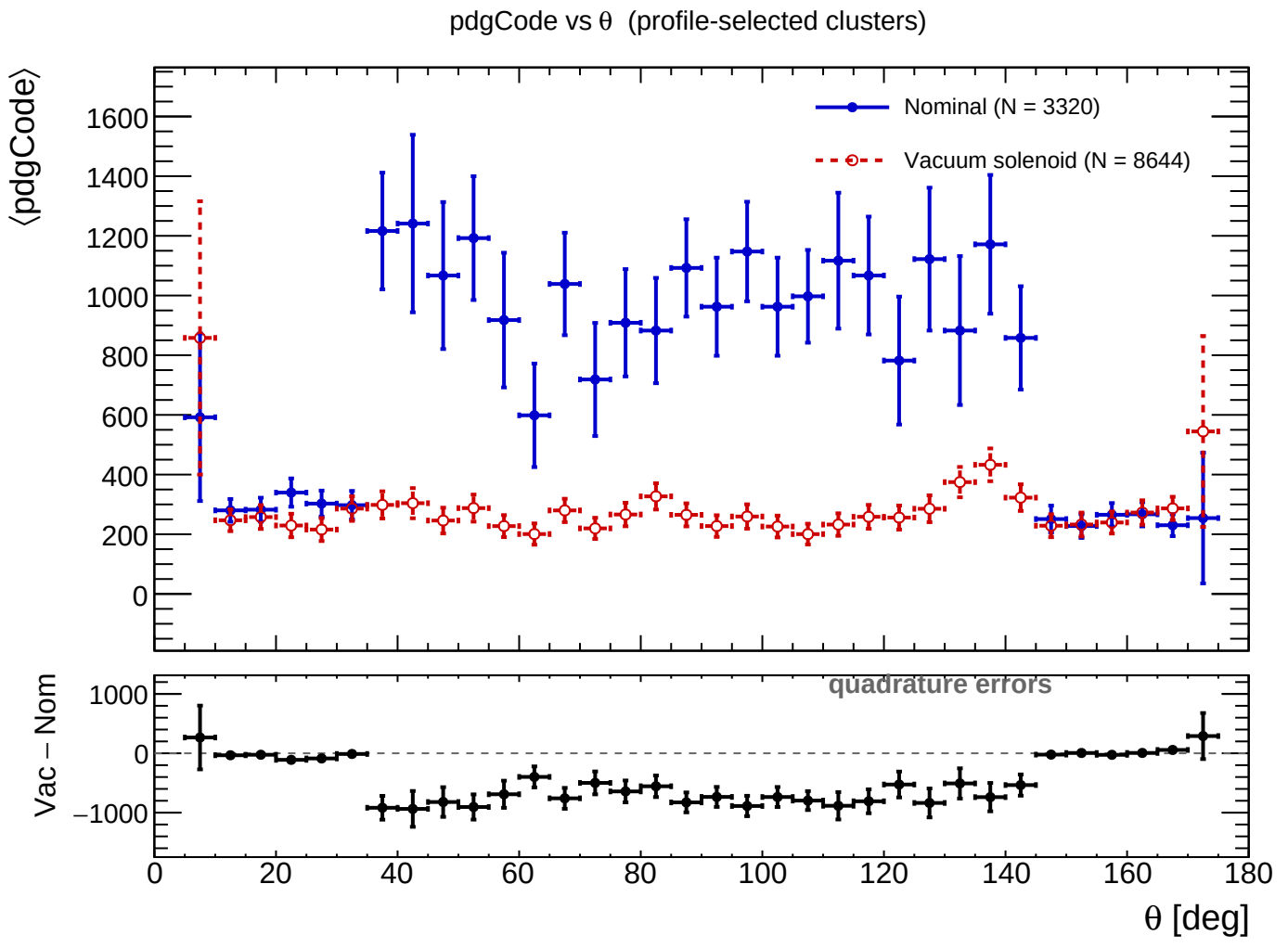

Number of calo hits vs θ (all clusters)
⟨Number of calo hits⟩
Nominal (N = 11972)
Vacuum solenoid (N = 11873)
Vac – Nom
quadrature errors
θ [deg]

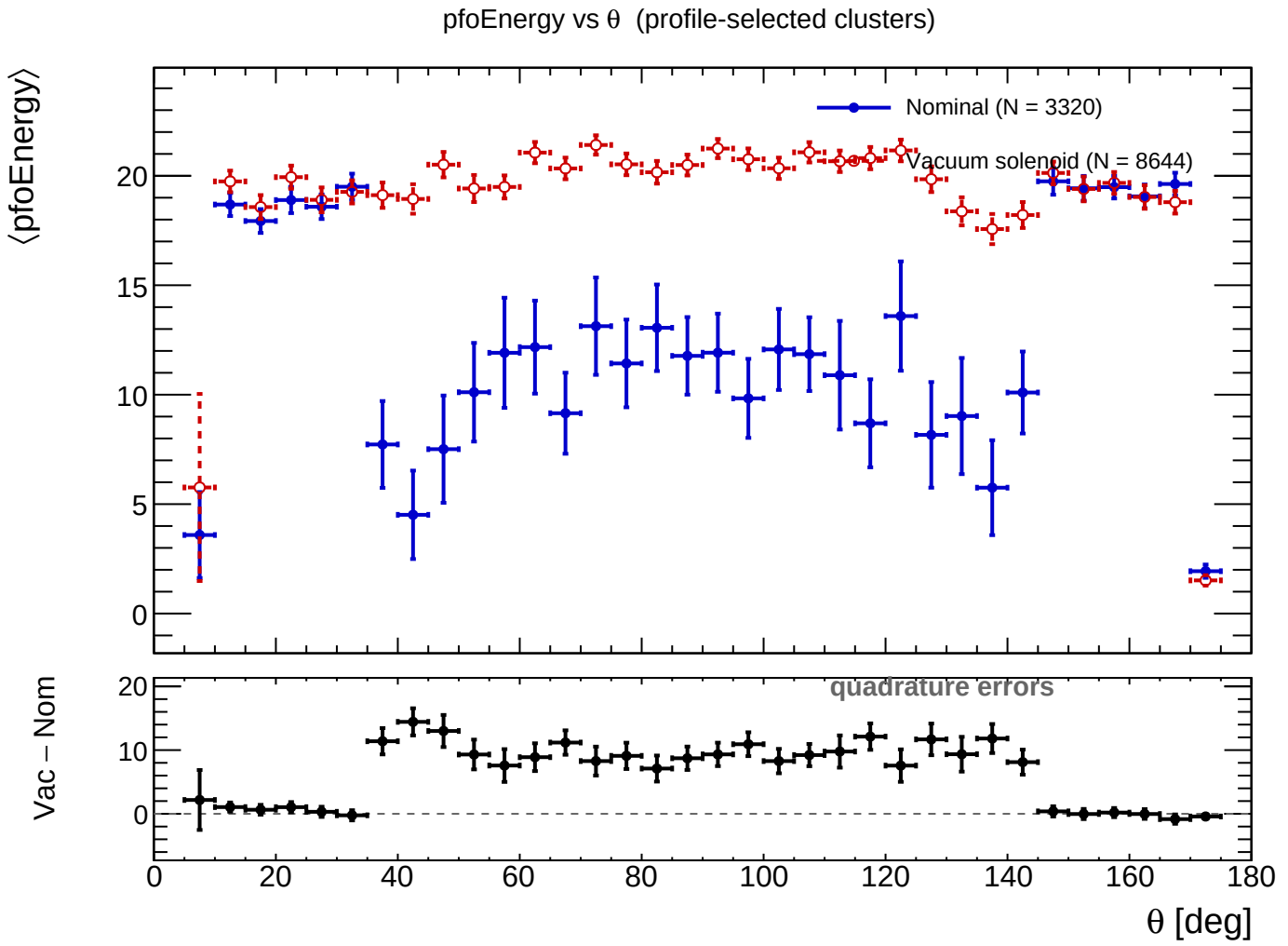

Inner pseudo-layer vs θ (profile-selected clusters)
⟨Inner pseudo-layer⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac – Nom
quadrature errors
θ [deg]

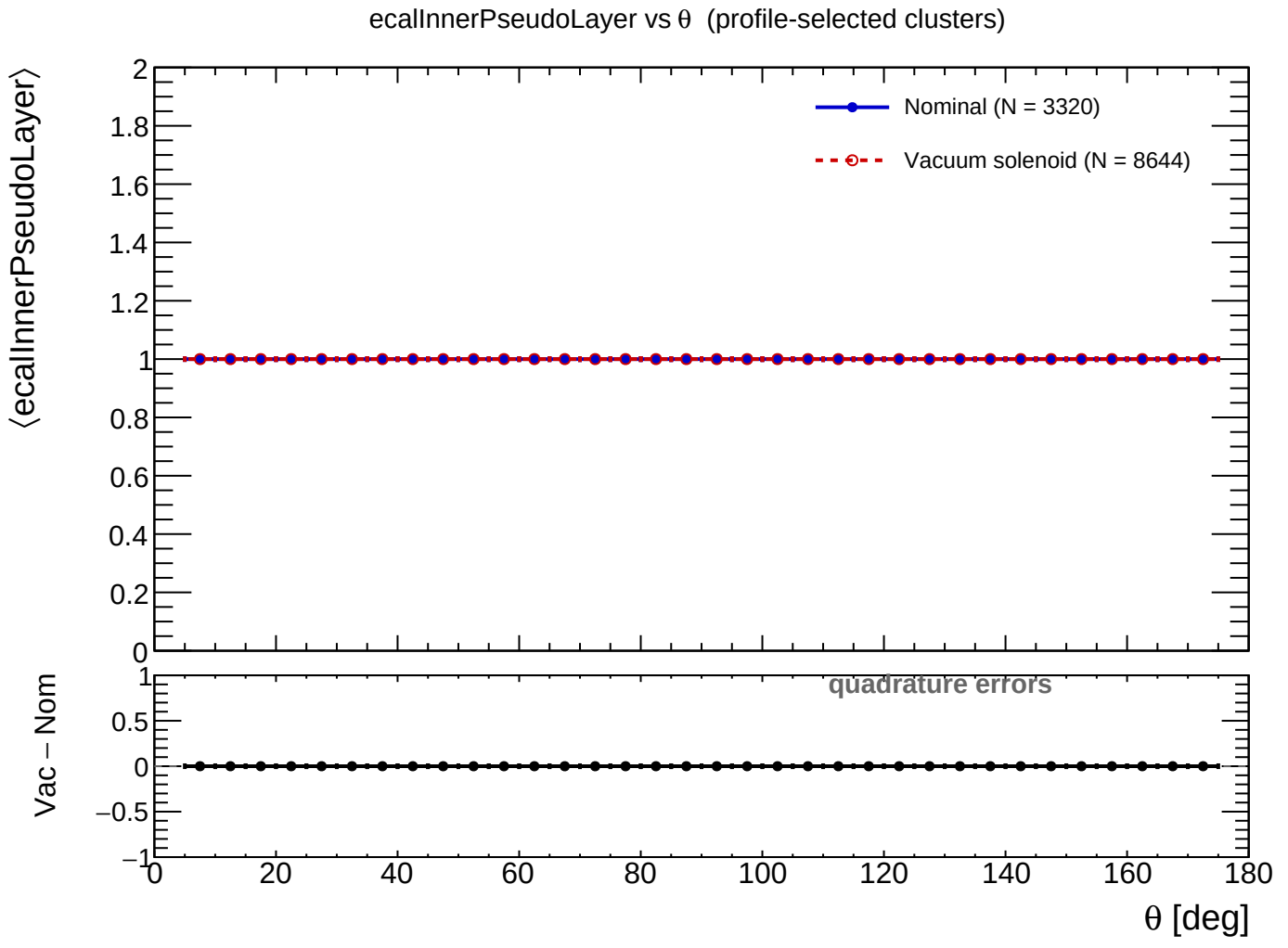

outerLayer vs θ (profile-selected clusters)
⟨outerLayer⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac − Nom
quadrature errors
θ [deg]

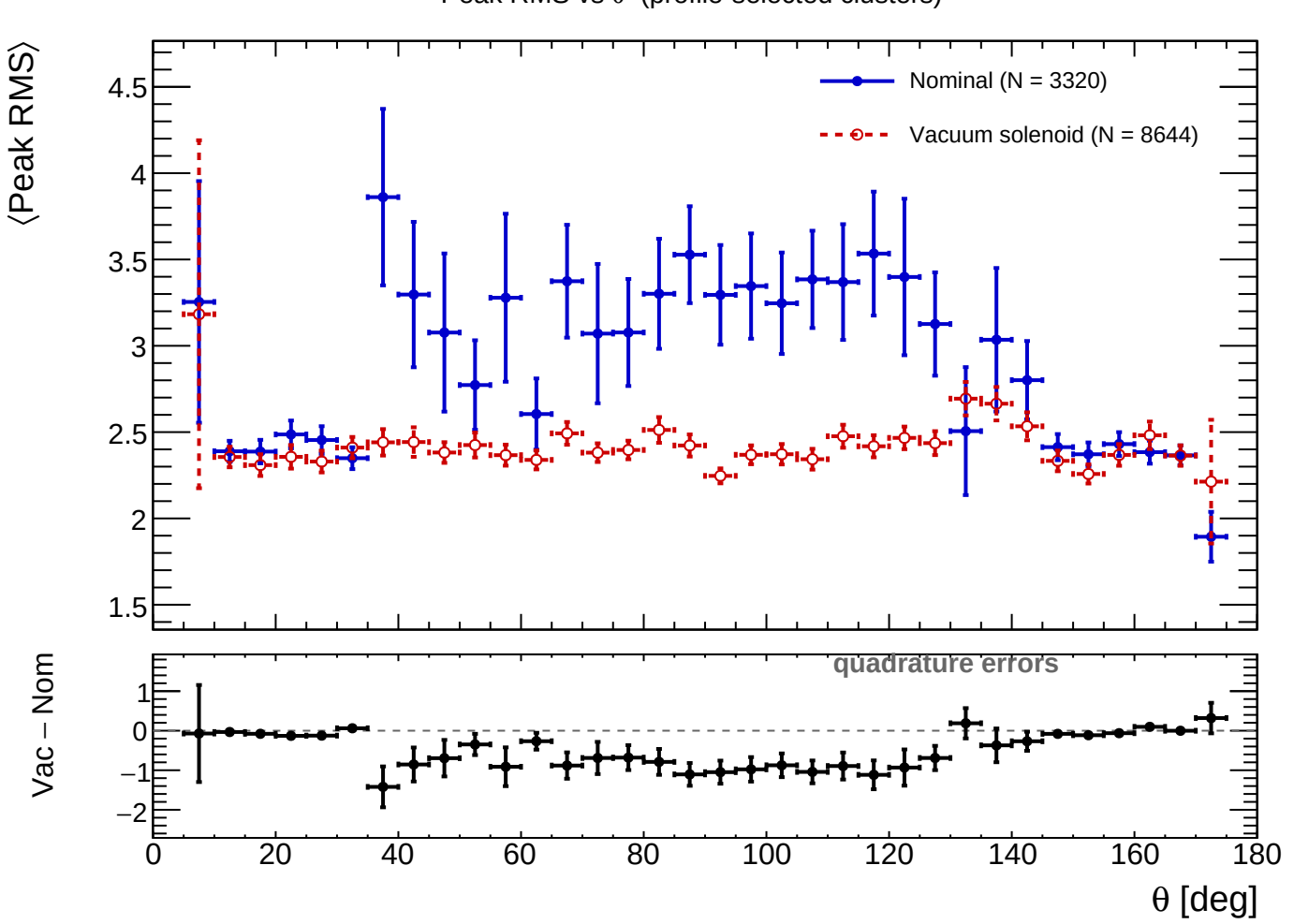

nLayers vs θ (profile-selected clusters)
⟨nLayers⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
35
30
25
20
15
Vac – Nom
quadrature errors
10
5
0
−5
0
20
40
60
80
100
120
140
160
180
θ [deg]

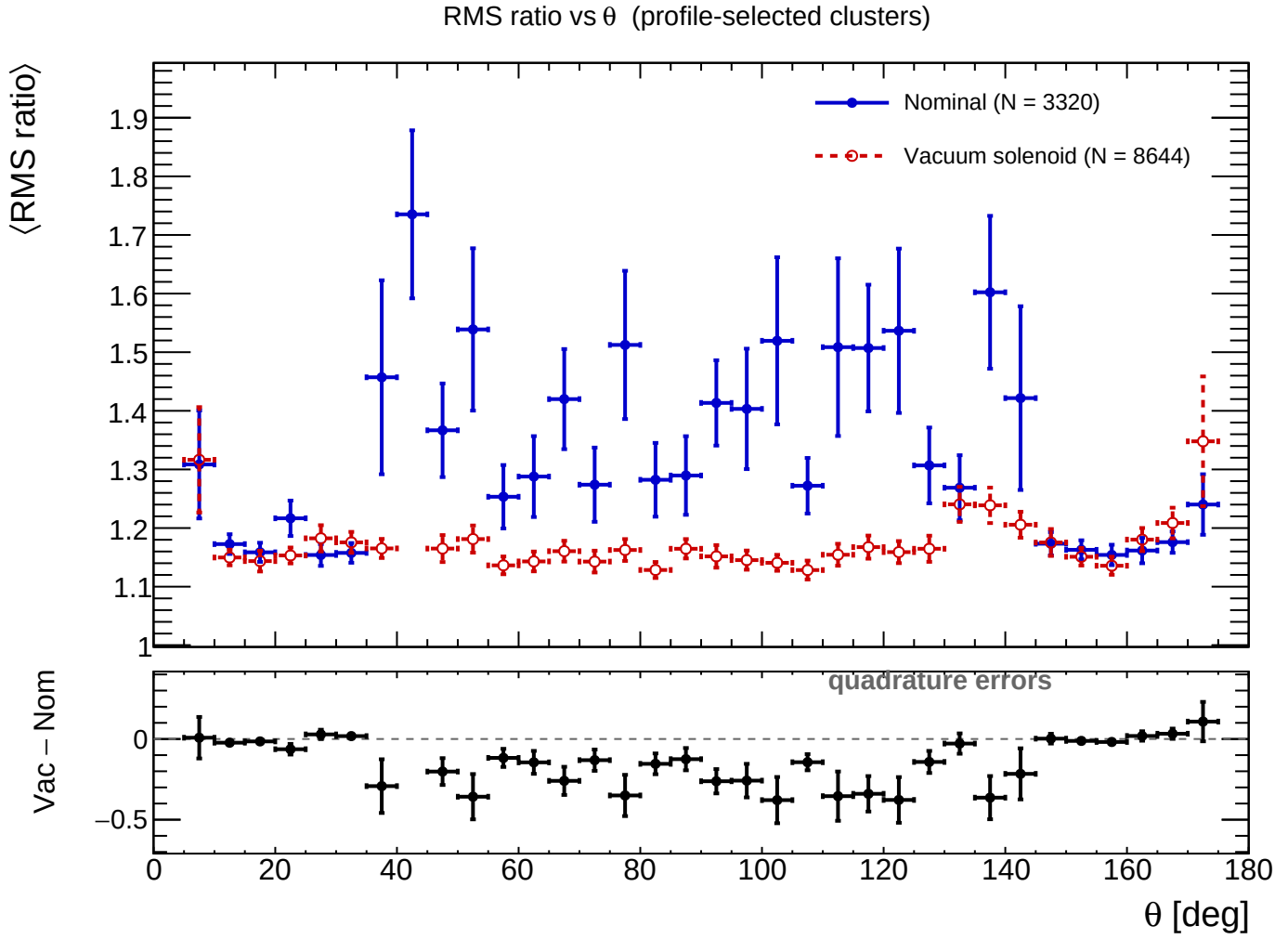

showerStartOffset vs θ (profile-selected clusters)
⟨showerStartOffset⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac − Nom
quadrature errors
θ [deg]

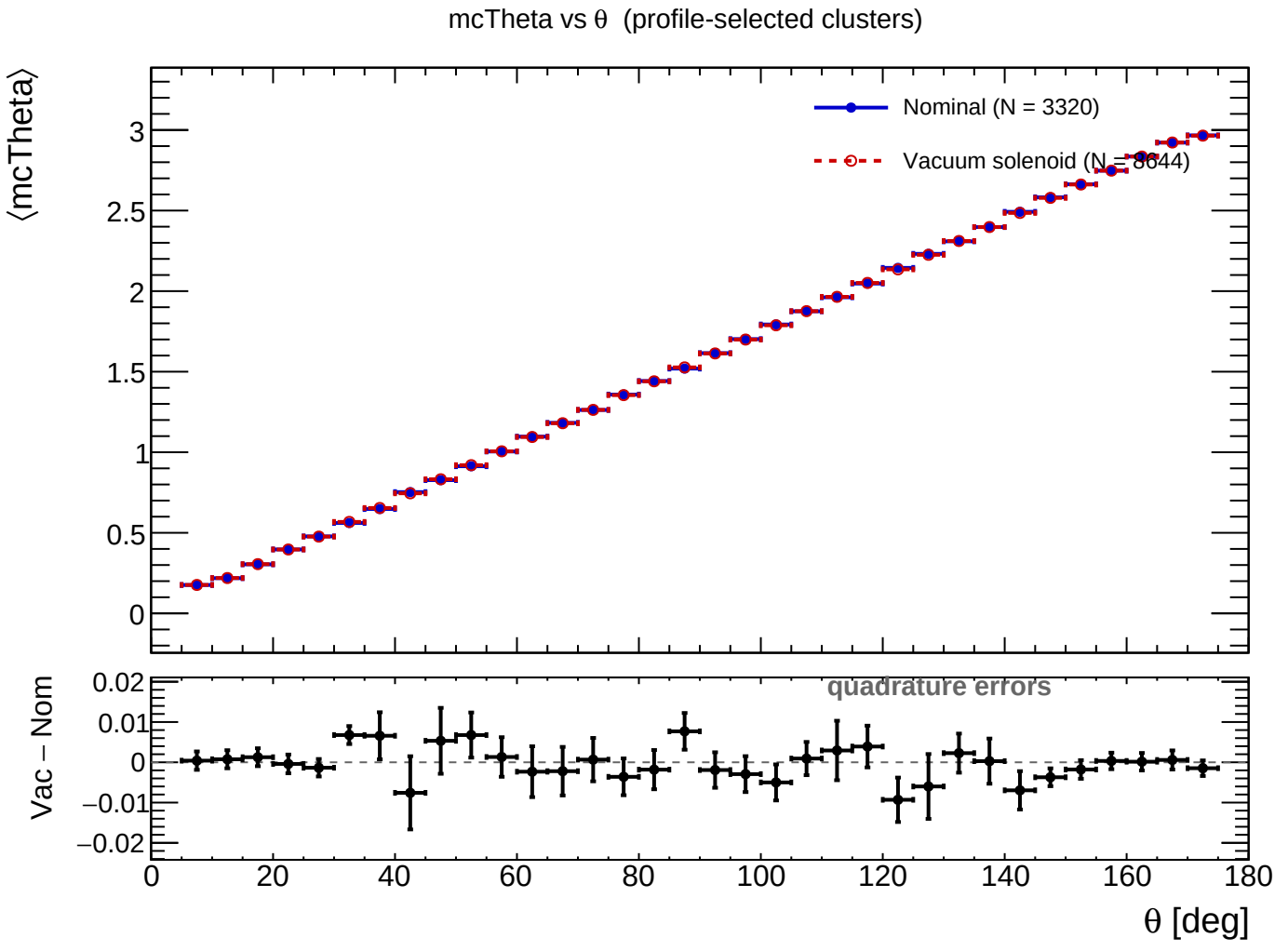

pdgCode vs θ (profile-selected clusters)
⟨pdgCode⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
1600
1400
1200
1000
800
600
400
200
0
Vac – Nom
quadrature errors
1000
0
−1000
0
20
40
60
80
100
120
140
160
180
θ [deg]

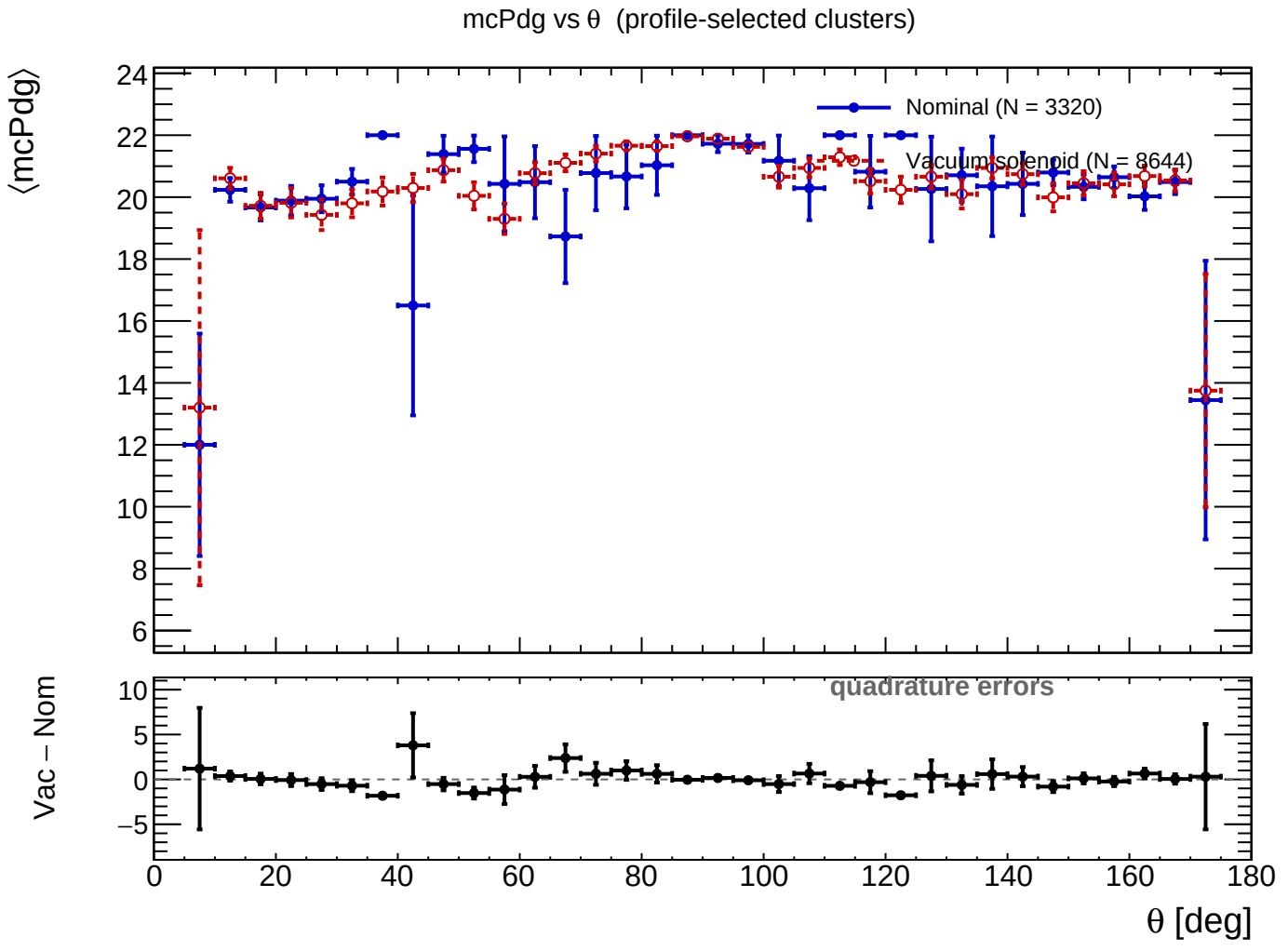

pfoEnergy vs θ (profile-selected clusters)
⟨pfoEnergy⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac – Nom
quadrature errors
θ [deg]


ecalInnerPseudoLayer vs θ (profile-selected clusters)
⟨ecalInnerPseudoLayer⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac − Nom
quadrature errors
θ [deg]


Peak RMS vs θ (profile-selected clusters)
⟨Peak RMS⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
4.5
4
3.5
3
2.5
2
1.5
Vac – Nom
quadrature errors
1
0
−1
−2
0
20
40
60
80
100
120
140
160
180
θ [deg]


RMS ratio vs θ (profile-selected clusters)
⟨RMS ratio⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
1
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9
Vac − Nom
quadrature errors
0
−0.5
0
20
40
60
80
100
120
140
160
180
θ [deg]


mcTheta vs θ (profile-selected clusters)
⟨mcTheta⟩
Nominal (N = 3320)
Vacuum solenoid (N = 3644)
Vac − Nom
quadrature errors
θ [deg]


mcPdg vs θ (profile-selected clusters)
⟨mcPdg⟩
Nominal (N = 3320)
Vacuum solenoid (N = 8644)
Vac – Nom
quadrature errors
θ [deg]